EL NIÑO ACTION BRIEFING · 29 SEPTEMBER 2026

South San Francisco, San Mateo County, California

Here El Niño brings water rather than heat: atmospheric rivers arriving at king tide back Colma Creek up into the flats east of US-101. The single most important step is to get storm-night beds to the people sleeping in cars and tents along the creek before 15 November.

IMPACT REGION	EXCESS HEAT DEATHS	WINDOW	DOMINANT HAZARD
San Mateo County, CA (inferred)	None published for the US, California, or the county	n/a here · global 6-month figure 239,000 ± 11,000	Flood: atmospheric river plus king tide; power outage second

16/16 recommendations checked. 14 corrected, 0 unverified, 1 contraindicated (a funded CO-alarm giveaway), 1 confirmed as proposed. Three proposed "do not do" items were also tested: two held, and one (sandbags) was downgraded from harmful to merely low priority. Exposure: Climate Impact Lab, *451,000 at Risk*, September 2026, pre-peer-review. It publishes **no figure for the United States, California, or San Mateo County**, and heat is not this place's hazard. No population-scaled figure is given, because scaling a heat number onto a flood hazard would be meaningless. The funding target fell 17.5% in verification, from $325k to $268k, which is within the one-third tolerance.

EXECUTIVE SUMMARY — AT A GLANCE



37.65°N 122.42°W, United States of America · Natural Earth

- **The risk.** Atmospheric rivers arriving at king tide back Colma Creek up into the flats east of US-101, where people sleep in cars and tents along the creek.
- **First action.** The County Human Services Agency, Samaritan House, LifeMoves, and the City get storm-night beds to people along Colma Creek and remove the weekday referral step by 15 November.
- **The money.** $77,000 floor, $268,000 target, $694,000 ceiling; the floor buys 10 storm nights for 20 people and 50 batteries.
- **Do not.** Do not clear creek-bank encampments in the weeks before the storms: it scatters people away from the outreach workers who could warn them.

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* Suggested formulation: check with a health worker before sharing.

WHAT THIS PLACE SHOULD DO RANKED

Ranked by expected lives saved per unit of effort before the peak. The rains are forecast to arrive late; NOAA's October–December signal favours only southern California, and the wet signal peaks in December–April. Backing lenses are shown on each action, and the RACI chart below assigns each one. An action several lenses reached independently is the stronger bet, but only where its evidence came from somewhere like here.

1. Get storm-night beds to the people living in cars and tents along Colma Creek, and remove the weekday referral step before 15 November

PARTIALLY CONFIRMED | EMERGENCY MGR · PUBLIC HEALTH · FRONTLINE · SKEPTIC · PHILANTHROPIST | INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
County Human Services Agency, with Samaritan House, LifeMoves, and the City of SSF	Referral bypass agreed by 15 Nov; bridge outreach walks in November	Existing Inclement Weather Program, plus ~$148k philanthropic top-up (see funding)	Deaths from drowning and hypothermia among creek-bank and vehicle dwellers

The county's Inclement Weather Program runs 15 Nov–15 Apr and already activates on “high probability of rain or a significant weather system”, not only on freezing temperatures. The bottleneck is the entry route. It runs through a Core Service Agency and then Coordinated Entry, which operates **Monday–Friday, 9–5**, and storms arrive at night and on weekends. The capacity nearest the creek is Samaritan House's **Safe Harbor** shelter in SSF (105 beds plus 10 cots). The county's Event Center overflow in San Mateo holds **53 people**, for one to three days per activation. OneShoreline's September 2024 committee packet records encampments under the **S. Airport Blvd and Produce Ave bridges**, which sit inside the flood channel. The 2026 county count found 1,145 unsheltered people: **482 in cars, 311 in RVs, and 246 on the street**. The action: pre-authorise storm-night referrals for people on the creek and in vehicles; send outreach workers the camps already know to agree pickup points at the bridges in November; and secure written daytime-refuge commitments from parish halls and libraries, because Safe Harbor closes at 7am while an atmospheric river can last all day.

* Suggested formulation: check with a health worker before sharing.

Transferability: the evidence that cold-weather shelter reduces deaths is thin. The one systematic review (Akhanemhe et al., *Public Health*, 2025) found little outcome data. The ~40× hypothermia-hospitalisation ratio is KFF Health News's own inference, not a study result. The case rests on the direct mechanism: the flood channel is where people sleep. No SSF-level unsheltered count exists, and that is the largest unknown in this whole briefing.

2. Reach the 341 power-dependent Medicare patients in 94080 through Medical Baseline, home-health agencies, and CID, and fill the battery gap the utility programmes leave

PARTIALLY CONFIRMED | PUBLIC HEALTH · EMERGENCY MGR · ECONOMIST · PHILANTHROPIST | INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
CID San Mateo (PG&E DDAR partner), home-health agencies, SSF Fire	1 December	Enrolment: staff time. Batteries: ~$80k for 100 units at ~$800	Deaths from failed oxygen, ventilator, and dialysis equipment during outages

HHS emPOWER (pulled directly from its REST service, updated 21 Sep 2026) counts **341 Medicare beneficiaries in ZIP 94080 on electricity-dependent equipment**, out of 13,443. Of the 341, 72 also use oxygen and 173 use home health. The county total is 3,688. Named lists go only to the state health department, and only after an incident, so the county and the city cannot pull names before a storm. Outreach has to run through Medical Baseline, home-health providers, and CID. **The gap:** PG&E's Portable Battery Program requires at least one wildfire shutoff or three or more fire-safety outages since 2024. The DDAR programme requires a High Fire-Threat District or two or more shutoffs since 2020. Storm outages in urban SSF qualify for neither. A 1 kWh battery costs about $800 and runs an oxygen concentrator for only about 2.5–3 hours, so the battery is a bridge to a shelter with power, not a substitute for one.*

* Suggested formulation: check with a health worker before sharing.

Transferability: the mortality evidence (the 2003 New York blackout: accidental deaths +122%, disease deaths +25%, ~90 excess; Anderson & Bell, *Epidemiology* 2012) comes from an August heat-season outage, and much of that excess was heat. The part that transfers to a mild coastal winter is equipment failure. The count covers Medicare only and misses under-65 users.

3. Pre-plan the US-101 / Oyster Point closure and pump staging for the three king-tide windows, triggered at seven days' lead

PARTIALLY CONFIRMED | EMERGENCY MGR · HISTORIAN · PUBLIC HEALTH · SKEPTIC | INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
SSF Fire (the city has no separate OES), with Caltrans and CHP; OneShoreline advising	**Plan by 15 Nov; windows ~24–26 Nov, ~22–25 Dec, ~20–23 Jan**	**Staff time within existing budgets; no published figure**	**Vehicle drownings at flooded crossings, and a repeat of the unplanned 2022 closure**

On 31 December 2022 CHP closed US-101 in both directions at Oyster Point from about 11am because “water is not receding due to non-stop rainfall & high tides”. Northbound reopened about 8pm. **Cars and a big-rig then drove through the closure and got stuck.** Colma Creek's outlet is tidal: at high tide it cannot drain, so this is a tide problem as much as a rain problem. The trigger: whenever a CW3E forecast of an AR2-or-stronger atmospheric river (on the Center for Western Weather and Water Extremes' 1–5 scale) lands inside a king-tide window, pre-stage pumps and barricades, and have physical closure points staffed at the underpasses. The CW3E forecast runs seven days out, so a months-ahead cross-index is not possible. The plan should be written now and triggered then. Confirm exact tide times at NOAA station 9414290.

* Suggested formulation: check with a health worker before sharing.

Transferability: good for the mechanism. Nationally, 63% of flood deaths with known circumstances involve vehicles (Ashley & Ashley 2008, 1959–2005 data). No SSF drowning is on record; the 2022 event was a near-miss, not a fatality.

4. Run carbon-monoxide warnings in Spanish, Tagalog, and Chinese through parishes before the first outage, with "no charcoal or grills indoors" first

PARTIALLY CONFIRMED | PUBLIC HEALTH · SKEPTIC · FRONTLINE | INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
County Health with St. Veronica, All Souls, and Filipino and Chinese community groups; SSF CERT	Masses and meetings in October–November	Translation and printing; *unfunded*, small	CO deaths from charcoal and generators used indoors during outages

The best evidence comes from a close analogue: a mild Pacific windstorm, not an ice storm. After the 1993 Puget Sound windstorm, **half of CO patients did not speak English, and charcoal caused 65% of incidents against 24% for generators** (CDC MMWR 1993). The pattern repeated in Seattle in 2006. The same outage failure kills here: in January 2023, during a week-long outage at Sea Ranch, two men died from a generator running in a hallway. About 52% of SSF residents speak a language other than English at home. The message leads with charcoal and grills, then generators outdoors only.*

Transferability: strong. The setting matches on climate, outage type, and an immigrant population. The 1993 MMWR is the correct source; the lens first cited a different paper.

5. Warn the flats in English and Spanish by Wireless Emergency Alert, and in Tagalog and Chinese through parish phone trees; do not rely on opt-in SMC Alert

PARTIALLY CONFIRMED | SKEPTIC · FRONTLINE | INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
County OES (WEA); parishes and community groups (phone trees)	Phone-tree captains named by 15 Nov	Volunteer time; *unfunded*	Warnings that never reach limited-English elders east of 101 and in Lindenville

* Suggested formulation: check with a health worker before sharing.

In October 2019 only **10.6%** of county residents were enrolled in SMC Alert, ranging from 2.2% in East Palo Alto to 58.1% in Portola Valley; Daly City was 2.6% (Civil Grand Jury, August 2020). The county declined opt-out auto-enrolment, and the system is still opt-in. **WEA cannot carry Tagalog or Chinese this winter:** the FCC's multilingual template rule has a compliance date of 12 June 2028, so until then WEA free text is English and Spanish only. The phone tree fills that gap: one captain per pew or block, phoning elders who do not text.

Transferability, with a caution: at Pajaro in March 2023, bilingual crews went door to door the day before the levee breach, and 56 people still needed rescue. Warnings without a place to go and a way to get there are not enough, which is why this ranks below action 1.

6. Remove sediment from the Colma Creek channel and finish the outfall repairs before the first big storm

PARTIALLY CONFIRMED | EMERGENCY MGR · HISTORIAN | INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
County DPW and SSF crews under maintenance agreements with OneShoreline	**Before the first atmospheric river, subject to the permit work window**	**Colma Creek Flood Control Zone operations budget (property tax)**	**Lost channel capacity at tidal lock-out**

OneShoreline funds the work; SSF, Colma, and County DPW do it, with quarterly joint inspections. The permits cap sediment removal at 400 cubic yards a year, between the Caltrain bridge and San Mateo Ave. A separate project repairs 15 degraded outfalls between Utah Ave and US-101. The 2023 Schaaf & Wheeler condition assessment found settlement of the north bank between Orange Ave and the dissipater and recommended a geotechnical investigation, which is **not a fix that can be finished this autumn**. The channel was concreted between 1964 and 1978, and only the San Mateo Ave reach has been rebuilt since, in 2006.

* Suggested formulation: check with a health worker before sharing.

Feasibility check needed: the cited US Army Corps nationwide permit ran to 14 March 2026, so confirm it has been renewed. The in-channel work window under the state Fish and Wildlife agreement was not found.

7. Enrol eligible households in CARE, FERA, Medical Baseline, and HEAP before November

CONFIRMED | ECONOMIST · PUBLIC HEALTH | INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
YMCA Community Resource Center; Central Coast Energy Services (HEAP)	31 October	Outreach staff; benefits paid by PG&E ratepayers (CPUC) and federal LIHEAP	Heating being rationed or shut off in a cold, wet winter

CARE: at least 35% off electricity and 20% off gas. FERA: 18% off electricity. HEAP in San Mateo County is run by Central Coast Energy Services (1-888-728-3637) and pays once per calendar year. The CPUC already bars disconnection of Medical Baseline customers on payment plans. A statewide disconnection moratorium switches on automatically, but only when **the Governor or the President** declares an emergency; a county proclamation does not trigger it.

This is harm reduction, not a proven life-saver. It is ranked here because it is cheap, it is fully funded, and the enrolment windows close before the storms.

8. Write a pre-agreed trigger for bridge cash, so a county emergency proclamation releases money within days rather than ten weeks

PARTIALLY CONFIRMED | ECONOMIST · PHILANTHROPIST | INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
County with SVCF (a revived SMC Strong or a successor fund); YMCA CRC to disburse	Agreement signed by 1 December	$30k seed for SSF (see funding); vehicle *unfunded* until revived	Displaced households going without cash for weeks

* Suggested formulation: check with a health worker before sharing.

SMC Strong was a COVID fund launched in March 2020 with a $3M county seed. It shows no activity in 2025–26, so it has to be revived or replaced. Its small-business grants of up to $10k were run through SAMCEDA and the San Mateo Credit Union. The lens's "200 × $10k" precedent does not match the sources. The case for a trigger: Samaritan House's 2023 storm fund took applications from 22 March to 21 April for storms on 31 Dec–12 Jan, and awarded by lottery. That is roughly a ten-week lag. The YMCA Community Resource Center (1486 Huntington Ave) serves SSF, San Bruno, and Brisbane, and is the right channel.

In 2023, 236 displaced people were put in hotels across the whole county. Size the SSF pool to that scale, not larger. It should be written as bridge cash that arrives before any FEMA declaration, not as a replacement for FEMA.

9. Run a renters' contents-only flood insurance drive that closes 31 October

PARTIALLY CONFIRMED | ECONOMIST | INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
City of SSF with the YMCA CRC; households pay	Bound by 31 October (30-day wait)	Premium per quote; low-income subsidy *unfunded*	Uninsured loss (harm, not deaths)

As of 31 May 2026, about **190** NFIP policies were in force in ZIP 94080, only 9 of them contents-only, against about 3,960 countywide (OpenFEMA, re-queried). In the 2022–23 storms, NFIP paid **3 claims in 94080, totalling $56,372**. The standard 30-day wait applies, with no exception for renters, so a policy bought after about 1 December misses early-January storms. SSF has **no Community Rating System discount**. The Class 8 (10%) discount belongs to unincorporated San Mateo County, not the city.

* Suggested formulation: check with a health worker before sharing.

Ranked low on lives saved, but it is the only instrument here that pays out by contract.

10. List the homes at the foot of San Bruno Mountain's south slope and Sign Hill, and watch the USGS soil-moisture data

PARTIALLY CONFIRMED | HISTORIAN · EMERGENCY MGR | INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
SSF Public Works	30 November	Staff time	Debris-flow deaths, a low-probability risk here

San Mateo County took the worst El Niño landslide damage in the Bay Area in 1997–98: about **$55M** ($45M to structures and $10M to roads, USGS MF-2325-H), and the only death, at Loma Mar. But **SSF's own recorded risk is modest**. The General Plan rates it medium, and near San Bruno Mountain the 1998 damage was mostly cracked retaining walls. The damaging slides were in Pacifica, La Honda, Moss Beach, Daly City, and Portola Valley. The NWS–USGS debris-flow warning system ran from 1986 to 1995 and was never restored. USGS now publishes soil moisture but issues no warnings.

Demoted: the lenses reached for the county's landslide record, but most of that record comes from elsewhere in the county. Rainfall-threshold warning zones belong to Pacifica and the coastside.

11. For 2027–28, not this winter: price parametric flood cover for the Colma Creek zone, ask NWS to set flood stages at gauge MXSC1, and join the CRS

PARTIALLY CONFIRMED | ECONOMIST | INVENTORY: NOT CONSULTED

OWNER	BY WHEN	COST	PREVENTS
OneShoreline Board; City of SSF (CRS)	Start now; bind for 2027–28	Broker quote; payable from zone reserves if counsel agrees	Post-flood cash delay for the city

* Suggested formulation: check with a health worker before sharing.

The precedents are real. **Fremont** holds Floodbase cover that pays at least $200k when one contiguous flooded area exceeds 0.58 sq mi, measured by satellite. **Isleton** has a $2.5M FloodFlash policy funded by a $200k DWR grant; the verifier could not confirm it has bound. The Colma Creek zone held **$40.1M in reserves** at the end of FY25-26. But gauge MXSC1 has **no flood stages defined**, and Colma Creek flooding may never reach 0.58 sq mi of contiguous area. That basis risk has to be tested before any purchase. Procurement plus Board approval before December is not realistic.

Asking NWS for flood stages and joining the CRS both have negative net cost: premiums fall for every policyholder, and the city gains a gauge that future cover can be written on.

* Suggested formulation: check with a health worker before sharing.

WHO DOES WHAT

RACI CHART

R = Responsible (does the work) · **A** = Accountable (one per action; signs off and answers for it) · **C** = Consulted (asked before) · **I** = Informed (told after). Built from the owners named in the verified actions. No one here has been contacted or has agreed to a role; every row is a proposal.

Action	County Human Services Agency	Samaritan House & LifeMoves	City of SSF (incl. Public Works)	SSF Fire & CERT	CID San Mateo & home-health agencies	PG&E	Caltrans & CHP	OneShoreline	County Health	Parishes & community groups	County OES	County DPW	YMCA CRC & Central Coast Energy Services	SVCF
1. Storm-night beds for Colma Creek dwellers	**A**	**R**	**R**							**C**				
2. Reach power-dependent patients; fill battery gap				**R**	**A/R**	**C**								
3. Pre-plan US-101 / Oyster Point king-tide closure				**A/R**			**R**	**C**						
4. Multilingual carbon-monoxide warnings through parishes				**R**					**A**	**R**				
5. WEA alerts plus parish phone trees										**R**	**A/R**			
6. Colma Creek sediment removal and outfall repairs			**R**					**A**				**R**		
7. Enrol households in CARE, FERA, Medical Baseline, HEAP						**I**							**A/R**	
8. Pre-agreed trigger for bridge cash													**R**	**A**
9. Renters' contents-only flood insurance drive			**A**										**R**	
10. List homes below San Bruno Mountain			**A/R**											
11. Parametric cover, MXSC1 flood stages, CRS (2027–28)			**R**					**A**						

What the chart shows. **The City of SSF and SSF Fire between them touch eight of the eleven rows.** The city has no separate OES, so SSF Fire is both the emergency planner and a frontline responder, and that is the plan's biggest dependency. **Row 8 names “the County” but no county agency.** The chart puts the A on SVCF, which would hold the fund, and the vehicle itself, SMC Strong, is dormant and unfunded until it is revived. **Volunteers carry the language gap.** Rows 4 and 5 rest on parishes and community groups with unfunded translation and volunteer time, because WEA cannot carry Tagalog or Chinese until 2028. **Row 11 also depends on NWS, which no one here can direct.** Setting flood stages at gauge MXSC1 is an ask, not a decision the city or OneShoreline controls.

WHERE MONEY DOES THE MOST

FUNDING

This is the Philanthropist lens, reconciled against the ranked actions and verified. Placements are ranked by marginal lives saved per dollar. The envelope is bounded by what this place can spend before the peak, not by what it needs.

* Suggested formulation: check with a health worker before sharing.

FLOOR	TARGET	CEILING	LARGEST UNCERTAINTY
$77,000	$268,000	$694,000	Whether the shelter top-up adds beds or duplicates county-funded rooms

Floor: 20 people × 10 storm nights × $185 = $37,000, plus 50 batteries × $800 = $40,000. **Target:** 40 × 20 × $185 = $148,000; 100 batteries × $800 = $80,000; 100 kits × $100 = $10,000; 30 households × $1,000 = $30,000; CO alarms $0. Total $268,000. **Ceiling:** 80 × 30 × $185 = $444,000; 200 × $800 = $160,000; 80 × $1,000 = $80,000; 100 kits × $100 = $10,000. Total $694,000. The $185 is the county's 2014 motel rate of $135, inflation-adjusted, because no current rate is published. It is an estimate, not a quote. The target fell from the lens's $325k by 17.5%, less than a third. The verifier's ceiling of $714k counted 300 Meals on Wheels clients across five North County cities; this report counts SSF's ~100 only.

RECIPIENT	USE	AMOUNT	MECHANISM	BASIS	VERDICT
Samaritan House (Safe Harbor, SSF) and LifeMoves (county motel-voucher contract)	Action 1: storm-night rooms and outreach for Colma Creek and vehicle dwellers	$148,000	Top-up to the Inclement Weather Program	40 people × 20 storm nights × $185 (estimated). No published cost per death averted. For scale, 160 people were enrolled countywide in 2023	PARTIALLY CONFIRMED
CID San Mateo (PG&E DDAR partner)	Action 2: backup batteries for power-dependent residents outside utility eligibility	$80,000	Pre-positioned stock, via direct grant	100 units × ~$800 (Jackery 1000 v2 list price); about 29% of the 341 in 94080. Whether CID can deliver batteries bought with outside money is unconfirmed; call first	PARTIALLY CONFIRMED
YMCA Community Resource Center, 1486 Huntington Ave	Action 8: bridge cash for flood-displaced households before any FEMA declaration	$30,000	Unconditional cash transfer	30 households × $1,000. Cut from 100 because 236 people were displaced countywide in 2023	PARTIALLY CONFIRMED

* Suggested formulation: check with a health worker before sharing.

RECIPIENT	USE	AMOUNT	MECHANISM	BASIS	VERDICT
Meals on Wheels San Francisco (serves North County, including SSF)	Storm kits and wellness checks for homebound seniors	$10,000	Direct grant	~100 SSF clients × $100 (kit cost unverified). The lens named a non-existent “Meals on Wheels San Mateo County”	PARTIALLY CONFIRMED
SSF Fire / CERT	1,000 battery CO alarms for renters	$0 (was $30,000)	Pre-positioned stock	Landlords have been required to fit alarms since 2011–13 (SB 183), and alarm-law studies show no significant effect on deaths. Moved to “do not do”	CONTRAINDICATED

The funder's one thing: commit before 15 November, as standing top-ups to contracts that already exist. Once a federal declaration lands, FEMA and county money arrive and duplicate private dollars, so those dollars do the least then. The cheapest life saved is probably a person on backup oxygen whom no utility programme covers, because the utility's battery rules were written for wildfire, not rain.

DO NOT DO THIS — CONTRAINDICATED

Advice that circulates for this hazard and would cost lives here, or waste the window.

Clearing creek-bank encampments in the weeks before the storms

CONTRAINDICATED — PRINCIPLE HOLDS; NO SSF CLEARANCE DOCUMENTED

Moving people off Colma Creek before a storm scatters them away from the outreach workers who could warn them and bring them in. On 3 January 2023 San José had to order creek-dwellers out by loudspeaker and hailing device. The same month in Sacramento, two unhoused people were killed when trees fell on their tents. The verifier found no primary source for the claim that San José swept the camps as the storm arrived, and no planned Colma Creek clearance. Check SSF council agendas and the channel maintenance schedule anyway, because sediment work in the creek (action 6) could prompt one.

* Suggested formulation: check with a health worker before sharing.

Instead: outreach workers the camps already know agree pickup points in November and move people to a real bed when an atmospheric river is forecast.

Spending the preparedness budget on a mass CO-alarm giveaway *

CONTRAINDICATED AS A USE OF FUNDS

California has required CO alarms in single-family homes since July 2011 and in multifamily units since January 2013, and landlords must keep them working. The national study of state alarm laws (Hampson, *Clinical Toxicology* 2023) found no protective effect on deaths. New York City's law raised detection of CO hazards but produced no significant drop in deaths or hospitalisations. The storm-time killer here is fuel burned indoors, and an alarm only helps after that choice has been made.

Instead: put the money and the parish time into the multilingual “no charcoal indoors, generators outside” message (action 4). Hand out alarms only where a landlord has not complied.

Tested and not contraindicated: sandbag giveaways. The Skeptic called them harmful, but the Pitt Review (UK, 2008) found only that they were not “particularly effective” and said they “should not simply be withdrawn”. The county issued ~56,000 in 2023. They are low priority, not harmful. The better headline is crews placing barriers for older and disabled residents. **Not separately verified:** heat-framed El Niño messaging. It is wrong for a fog-cooled city whose forecast signal is rain, but the verifiers did not test it as its own claim.

WHAT HAS HAPPENED HERE BEFORE — THE RECORD

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
1964–78	Colma Creek Flood Control Zone created (1964); the creek is concreted	—	The channel was sized for rain in the 1960s, not for rain arriving at high tide
Jan 1982	Bay Area storm, before the 1982–83 El Niño (an analogue, not an El Niño event)	33 deaths in California, 25 of them in Bay Area landslides; $5M in damage in neighbouring Pacifica	The NWS–USGS debris-flow warning system, 1986; **switched off at the end of 1995**

* Suggested formulation: check with a health worker before sharing.

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
1997–98	Very strong El Niño; storms of 2–3 Feb 1998; Bay water ~2 ft above predicted tide on 3 Feb	~$55M in San Mateo County landslide damage, highest of the 10 Bay counties; 1 death (Loma Mar). About 17 Bay Area deaths (secondary source)	FEMA declaration for all 10 counties; the warning system was not restored
2006	San Mateo Ave reach of Colma Creek rebuilt, bridge raised	—	The only channel reach upgraded since 1978
2015–16	Strong El Niño, weak rain in Northern California	No SSF impacts found	A strong El Niño does not guarantee a wet winter here, but this year's forecast is wet
2019–20	OneShoreline created (AB 825)	—	The 2018 corridor-widening plan was moved into Lower Colma Creek planning, still at the public-input stage
31 Dec 2022 – Jan 2023	Atmospheric-river series (DR-4683); US-101 closed at Oyster Point by rain plus high tide	No SSF deaths found. County: 1,417 emergency calls on 31 Dec, 236 displaced people in hotels, 160 people in the Inclement Weather Program. NFIP paid 3 claims ($56,372) in 94080	2023 channel condition assessment; Event Center overflow shelter from Dec 2024

The recurrence nobody has noticed: the warning system built after 1982 was switched off two winters before San Mateo County took the worst El Niño landslide damage in the Bay Area. The flood fix proposed in 2018 is still in planning as the strongest El Niño on record may be arriving. The one upgrade built since, the Orange Memorial Park stormwater capture project (2022), cleans water; it does not add flood capacity. Each time, the protection was either retired or deferred just before the event it was meant for.

WHERE THE EXPERTS DISAGREE — UNRESOLVED

- **Better warnings save lives** (Emergency manager, Historian) versus **warnings are not the constraint** (Skeptic). On 31 Dec 2022 the closure was issued and drivers entered the water anyway, and at Pajaro people warned door to door still needed rescue. What would settle it:

* Suggested formulation: check with a health worker before sharing.

whether the people who stayed had somewhere to go. This report ranks shelter access above warning delivery on that basis.

- **Fund CO alarms** (Philanthropist) versus **alarm mandates show no mortality effect** (Public health, the verifier). The evidence sides with the second, and the placement was zeroed. A trial of targeted alarms in non-compliant rentals would settle it; none exists.
- **Sandbags are harmful theatre** (Skeptic) versus **the county issues them in bulk**. The verifier found low effectiveness, not harm.
- **Restore rainfall-threshold landslide warnings** (Historian) versus **SSF's landslide risk is modest** (the verifier). Right for the county, wrong for this city.
- **The insurance drive is a priority** (Economist) versus **it saves no lives**. It is kept, ranked low, because it is the only payout that is pre-arranged by contract.

WHAT NOBODY COVERED

BLIND SPOT

Nobody knows how many people sleep in the Colma Creek flood zone, and nobody checked care continuity when US-101 closes. The county count gives 1,145 unsheltered people but no SSF figure, so action 1 and the largest funding placement rest on a number no one has. Separately, 173 power-dependent patients in 94080 rely on home-health visits, and dialysis and home-care schedules were not examined for a day when the freeway is shut. The Public health lens's warning applies: most deaths in an outage winter will not be recorded as storm deaths. They will be missed treatments and cardiac events, visible only as all-cause excess mortality from December to March. An SSF street-and-vehicle count this autumn, and a call to the home-health agencies, would change the ranking more than any other two pieces of information.

HOW TO READ THIS

METHOD

The panel is author-built. Seven simulated expert lenses generated these recommendations. Where they agree, that is a strong hypothesis, not a consensus of the field. Every recommendation was then independently checked for evidence, transferability to this location, and feasibility before the peak. The verdict tags carry the result.

Mortality figures are resolved at Climate Impact Lab impact-region scale, roughly 300,000 people; they are not neighbourhood figures. The El Niño projection covers *heat only*. Drought, flood, fire, smoke, and crop failure are additional to it and are carried here by the historical record rather than by a published number. For South San Francisco that means the whole of the hazard is carried by the record.

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- **Scope.** The location was taken as given: South San Francisco, San Mateo County, California. Climate Impact Lab impact region, as the report records it: San Mateo County, CA, *inferred*, not read from the tool. The Impact Map's region lookup is a JavaScript application that could not be driven, and the public US data download is state-level only. Mortality figures exist only at that scale (impact regions average roughly 300,000 people); sub-city detail is used for hazard exposure and action design, not for death counts. Where administrative data stands in for the city, the unit is named: ZIP 94080 for emPOWER and NFIP; a Caltrans/C/CAG study area for some language figures.
- **Exposure.** Climate Impact Lab, *451,000 at Risk* (September 2026, pre-peer-review): ECMWF SEAS5 seasonal forecasts (51-member ensemble) applied to temperature–mortality relationships across 24,378 regions, against a 1996–2025 baseline, for June 2026 – February 2027. It covers heat only. It is never added to the Impact Map's long-run projections or quoted from the map's GDP-denominated mortality. The report PDF names no US, Californian, or North American figure. Heat is not the dominant hazard here, so the historical record and NOAA CPC's September 2026 outlook carry the hazard instead of a borrowed number. The outlook gives a >90% chance of a very strong event and a 50–60% chance of a wetter-than-normal December–April near the coast.
- **Expert panel.** Seven simulated expert lenses (emergency manager, public health scientist, frontline responder, skeptic, economist, historian, and philanthropist) each researched this place independently and in parallel, from web sources, to a fixed brief. A separate scoping agent resolved the exposure first.
- **Reconciliation.** Duplicate actions were merged and backing lenses recorded. Convergence was tested for shared bias before it counted in an action's favour. The five-lens shelter action rests on local facts (bridge encampments, referral hours), not on a shared canonical study. The four-lens power-dependency action rests on a New York heat-season blackout, and is flagged as such. Contradictions were kept, not resolved by dropping a side. Actions were ranked by expected lives saved per unit of effort before the peak, and each was given a named payer and trigger or marked unfunded. The funding placements were reconciled against the ranked list; the CO-alarm placement was dropped and the envelope re-summed.
- **Verification.** Six verifier agents checked every recommendation on three tests: evidence (the primary source and its effect size), transferability (whether that evidence came from somewhere like here), and feasibility (whether it can be done here before the peak). Financial actions had their own verifier, which checked availability in this jurisdiction, a named payer, a trigger threshold, and whether each already exists; it re-queried OpenFEMA directly. The funding envelope had its own verifier, which checked for real recipients, published unit costs, and duplication, then re-summed.
- **Verdicts.** *Confirmed*: evidence, transferability, and feasibility all hold. *Partially confirmed*: holds after the corrections shown. *Unverified*: could not be checked to a primary source. *Contraindicated*: would do harm or waste the window here.

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- **Adaptation Inventory.** Not consulted. The page cannot be fetched (it renders empty until a country is chosen in its UI), and the “Download Inventory” link leads to a form asking for name, organisation, and email, with no direct file URL. Every action is therefore tagged “not consulted”, which is not the same as “no precedent found”.
- **Health wording.** Passages where a wording error could hurt someone (medical equipment in an outage, carbon monoxide, generators, and charcoal) are marked "* Suggested formulation: check with a health worker before sharing." They were checked against published sources, not by a health professional. They are suggestions, not medical advice, and anyone passing them on should confirm them with a doctor, pharmacist, nurse, or community health worker first. The marker was added on 2 November 2026.
- **Rules.** No cost is derived by dividing a published total, and no local figure is invented; where data is thin, the report says so and shows the containing geography's number with its arithmetic. The $185 motel rate is an inflation adjustment of a 2014 published rate and is labelled an estimate. Every source is listed below with its verification status.
- **Limits.** The panel and verifiers are AI research agents working from web-available sources, not human experts, and no one in this place was interviewed. Run-specific gaps:
  - The Colma Creek gauge page returned no data.
  - No county after-action report for the 2023 storms was found.
  - The OneShoreline September 2024 committee packet was read from search excerpts, not the full PDF.
  - The Pitt Review was quoted via UK parliamentary evidence, not the original report.
  - SMC Alert's current enrolment is unpublished.
  - No SSF-level unsheltered count exists.
  - None of the organisations named in the funding table was contacted.

References · status after verification
Exposure: CIL El Niño report PDF (checked; no US figure) · NOAA CPC ENSO discussion, 10 Sep 2026 (checked) · CPC seasonal outlook, 17 Sep 2026 (checked)
Shelter: SMC Inclement Weather Program (verified) · Samaritan House (verified) · County Event Center shelter (verified) · KFF Health News (verified; the ratio is its inference) · 2026 count (news report of county count) · OneShoreline CAC packet, Sep 2024 (search excerpt)
Power and CO: HHS emPOWER (queried directly) · emPOWER individual data rules (verified) · PG&E Portable Battery Program (verified) · PG&E DDAR (verified) · CID San Mateo (verified) · Anderson & Bell 2012 (verified) · CDC MMWR 1993 (verified) · Sea Ranch deaths (verified) · Hampson 2023

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(verified) · Wheeler-Martin 2015 (verified) · SB 183 (verified)

Flood operations: ABC7, 31 Dec 2022 (verified) · NBC Bay Area (verified) · Ashley & Ashley 2008 (verified) · California King Tides (secondary; confirm at NOAA 9414290) · CW3E AR Scale (verified) · OneShoreline Colma Creek (verified) · County 2023 storm numbers (verified) · SSF Fire emergency preparedness (verified)

Warnings and landslides: Grand Jury on SMC Alert (verified) · FCC multilingual WEA compliance date (verified) · Pajaro 2023 (verified) · USGS MF-2325-H (verified) · USGS FS 98-089 (checked; preliminary $45.4M) · USGS Circ. 1283 (warning system) (verified) · SSF General Plan hazards (verified) · January 1982 floods (secondary) · 1998 Bay Area deaths (secondary)

Finance: OpenFEMA NFIP policies and claims (re-queried) · FEMA Community Status Book (verified) · Fremont parametric (verified) · Isleton parametric (grant verified; binding unconfirmed) · NWPS gauge MXSC1 (verified: no flood stages) · SMC Strong (verified; dormant) · Samaritan House 2023 storm fund (verified) · YMCA CRC (verified) · PG&E CARE/FERA (verified) · PG&E customer protections (verified) · Meals on Wheels SF, North County (verified) · SVCF disaster fund 2023 (verified)

Exposure data · Climate Impact Lab, 451,000 at Risk · Impact Map · Adaptation Inventory

Generated 29 September 2026 for South San Francisco, San Mateo County, California. El Niño Ready is volunteer-run and is not affiliated with Climate Week NYC, The Climate Group, or Columbia University.

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